June 2023 Hundred-Patch Benchmark

Metrics, plots, Taylor diagrams, optimization behavior, and runtimes

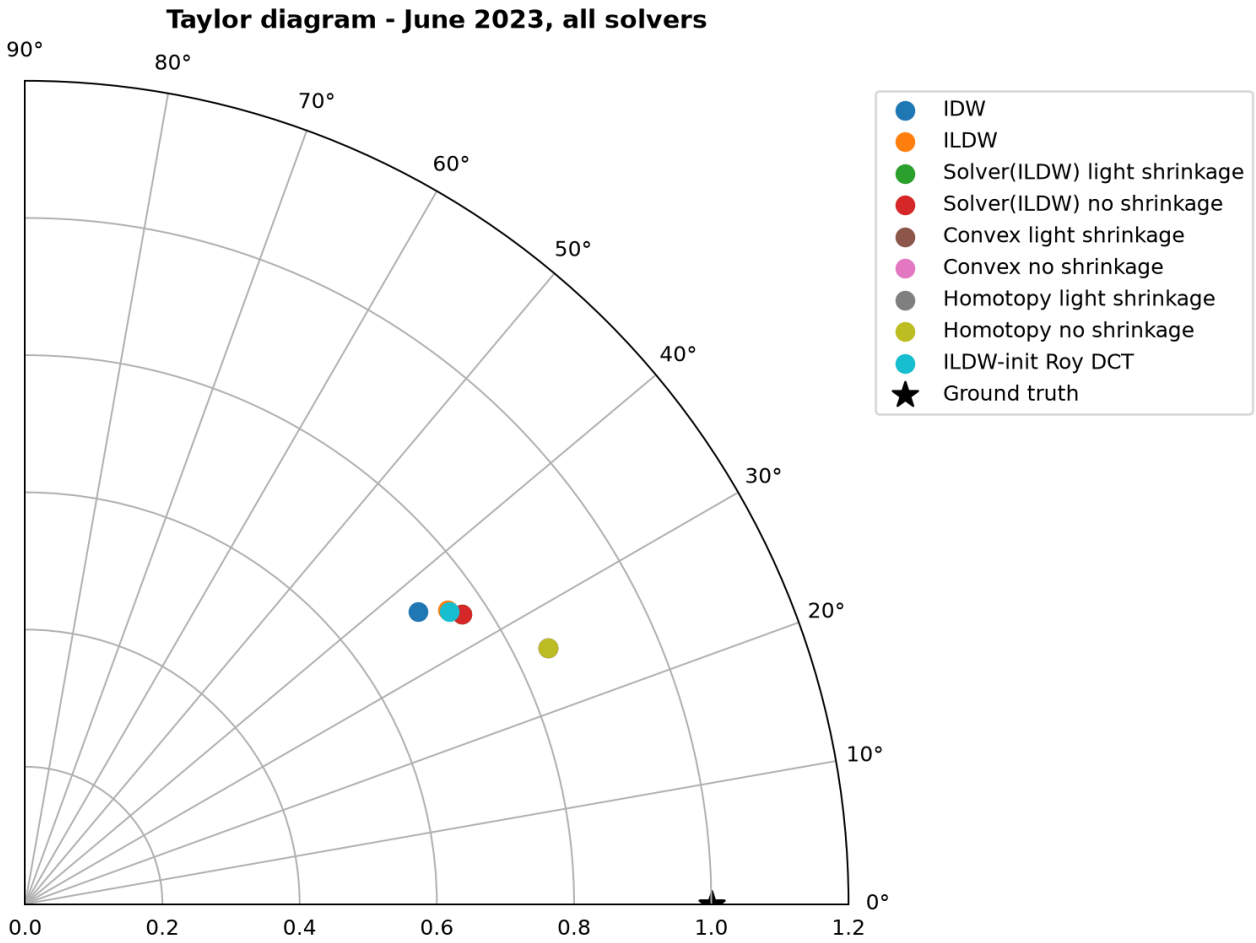
100 June 2023 patches. Supplementary Solver(ILDW) no-shrinkage, timed Solver(ILDW), and Roy DCT outputs are included from their complete June solution directories.

1. Summary metrics

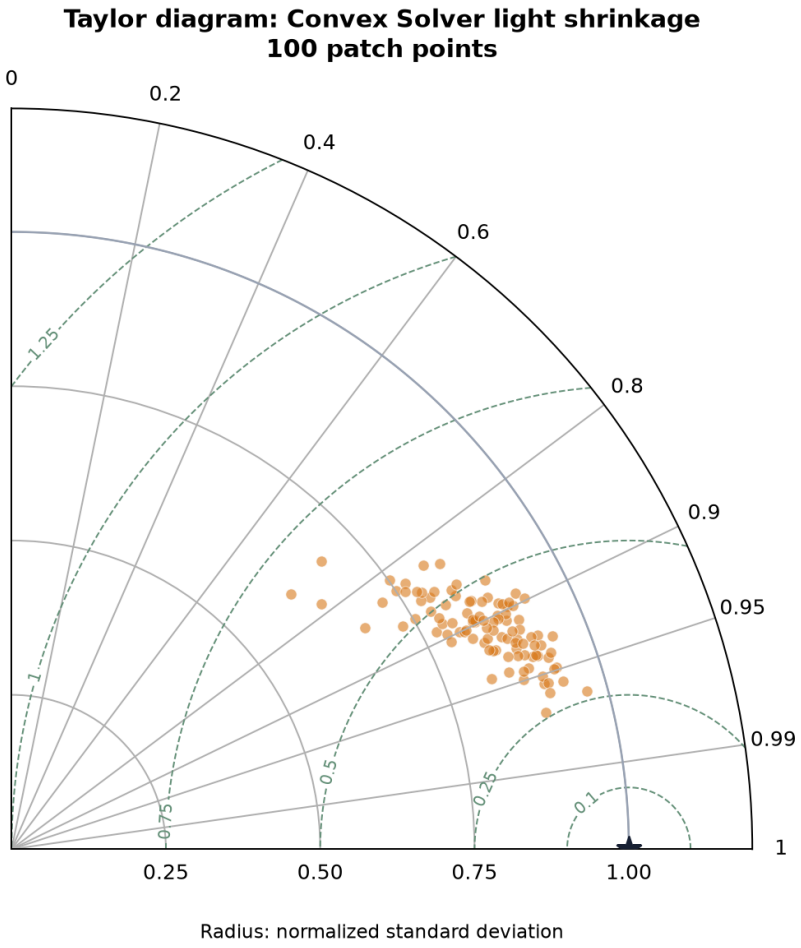
Solver	N	RMSE mean	RMSE min	RMSE max	Bias mean	Bias min	Bias max	Pearson mean	Pearson min	Pearson max
IDW	100	2.983	1.067	7.237	0.06152	-1.007	1.006	0.8056	0.6226	0.9347
ILDW	100	2.843	1.03	7.15	0.08054	-0.855	0.8596	0.8226	0.6294	0.9418
Convex Solver light shrinkage	100	2.161	0.8106	5.708	0.0697	-0.3811	0.5898	0.8982	0.732	0.9689
Convex Solver no shrinkage	100	2.161	0.8108	5.701	0.06996	-0.3865	0.5832	0.8982	0.7326	0.9685
Homotopy Solver light shrinkage	100	2.161	0.8106	5.707	0.06945	-0.3817	0.5886	0.8982	0.732	0.9689
Homotopy Solver no shrinkage	100	2.161	0.8109	5.701	0.06965	-0.3868	0.5825	0.8982	0.7326	0.9685
Solver(ILDW) light shrinkage	100	2.755	0.9785	7.147	0.0841	-0.8545	0.866	0.8342	0.6408	0.9595
Solver(ILDW) no shrinkage	100	2.76	0.9617	7.147	0.08137	-0.8554	0.8636	0.8346	0.6685	0.9515
ILDW-init Roy DCT	100	2.825	1.022	7.129	0.08265	-0.8434	0.8596	0.8253	0.6329	0.9432

Each row summarizes 100 patches. Mean/min/max are calculated separately across the per-patch RMSE, signed bias (prediction minus ground truth), and Pearson correlation. RMSE and bias are in mm/h. Only finite paired pixels are used; undefined correlations are excluded.

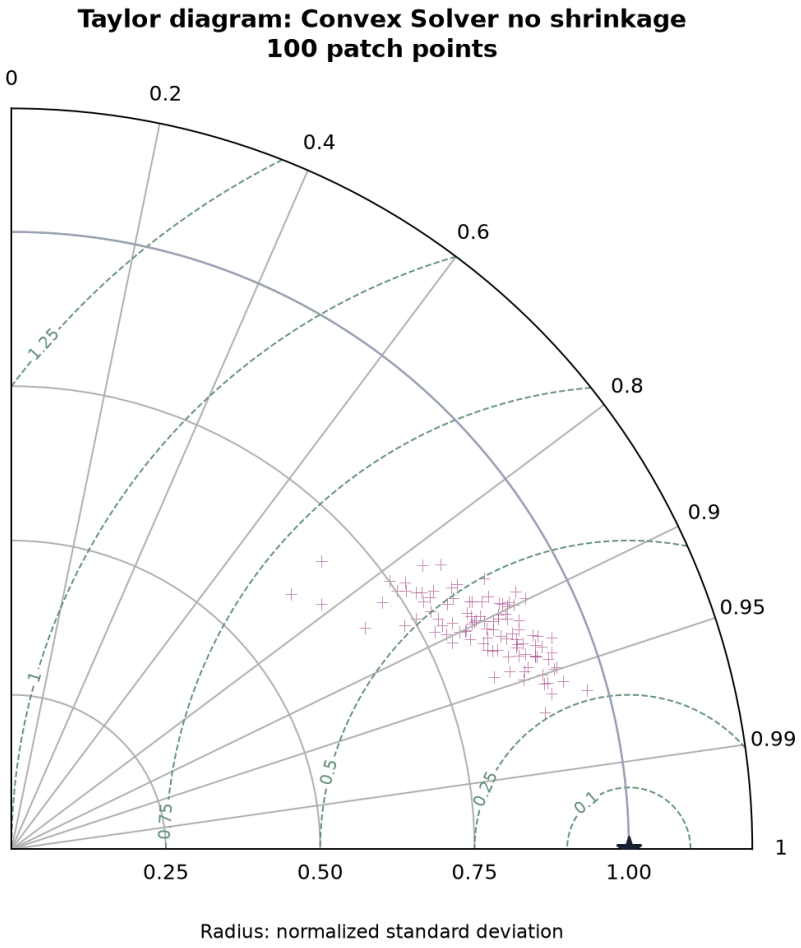
2. Taylor diagrams



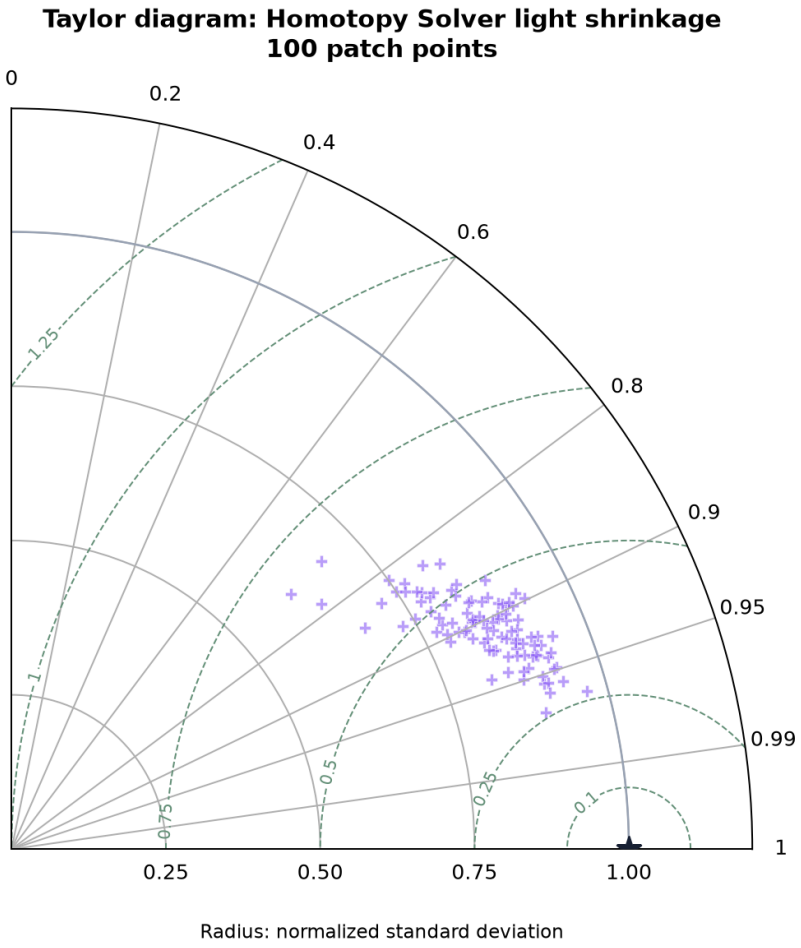
combined all solvers



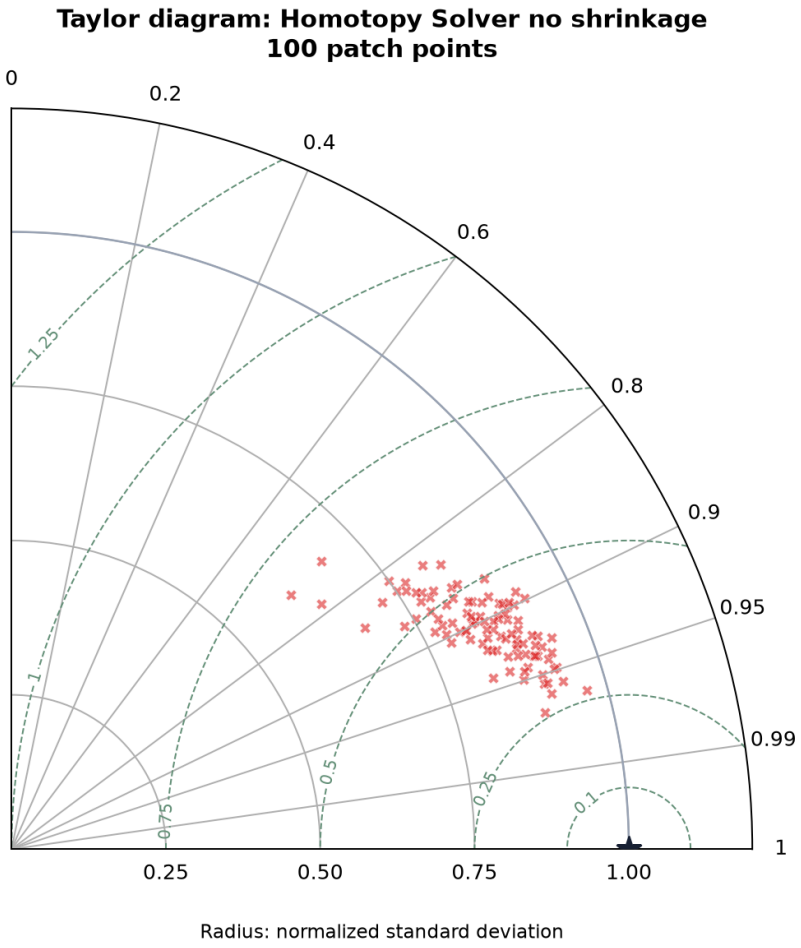
Convex Solver light shrinkage



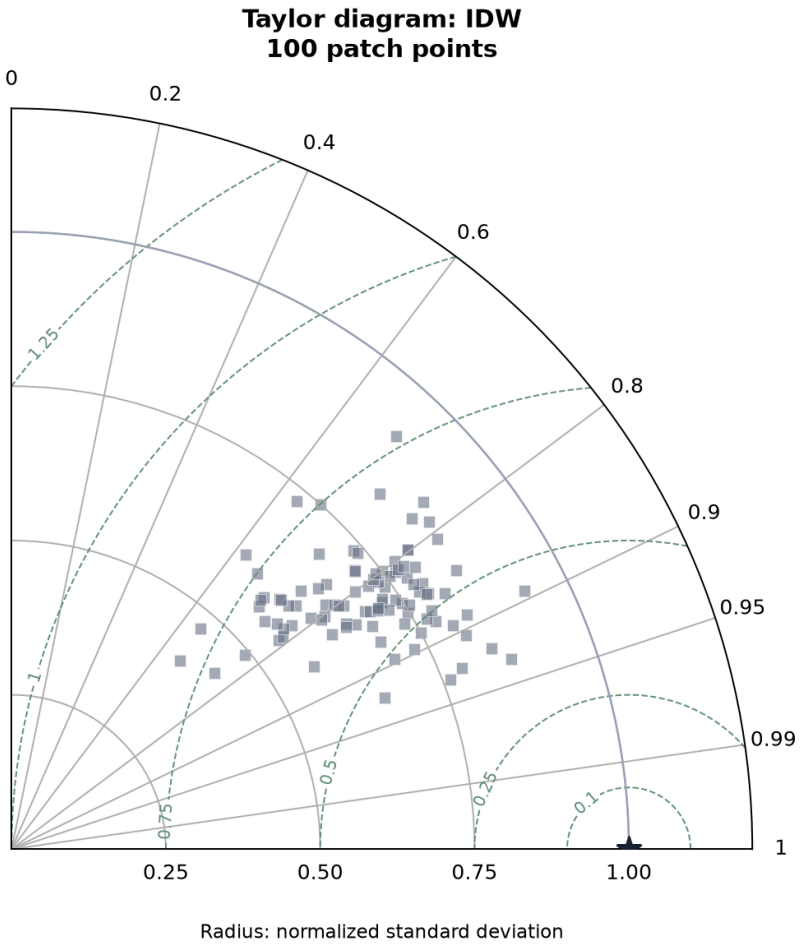
Convex Solver no shrinkage



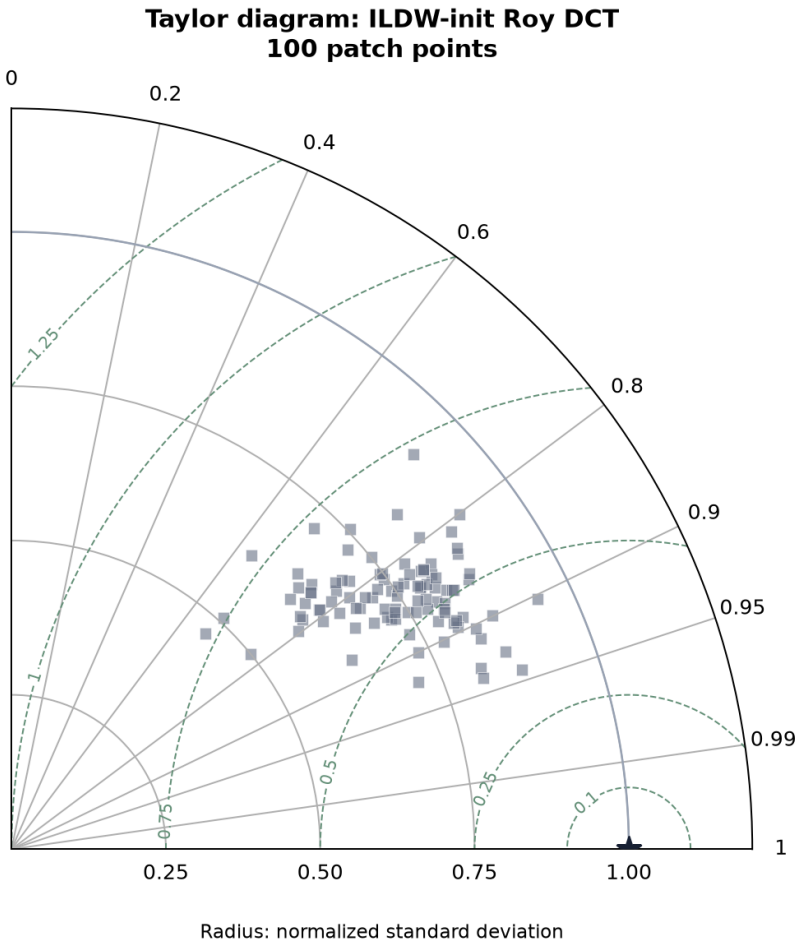
Homotopy Solver light shrinkage



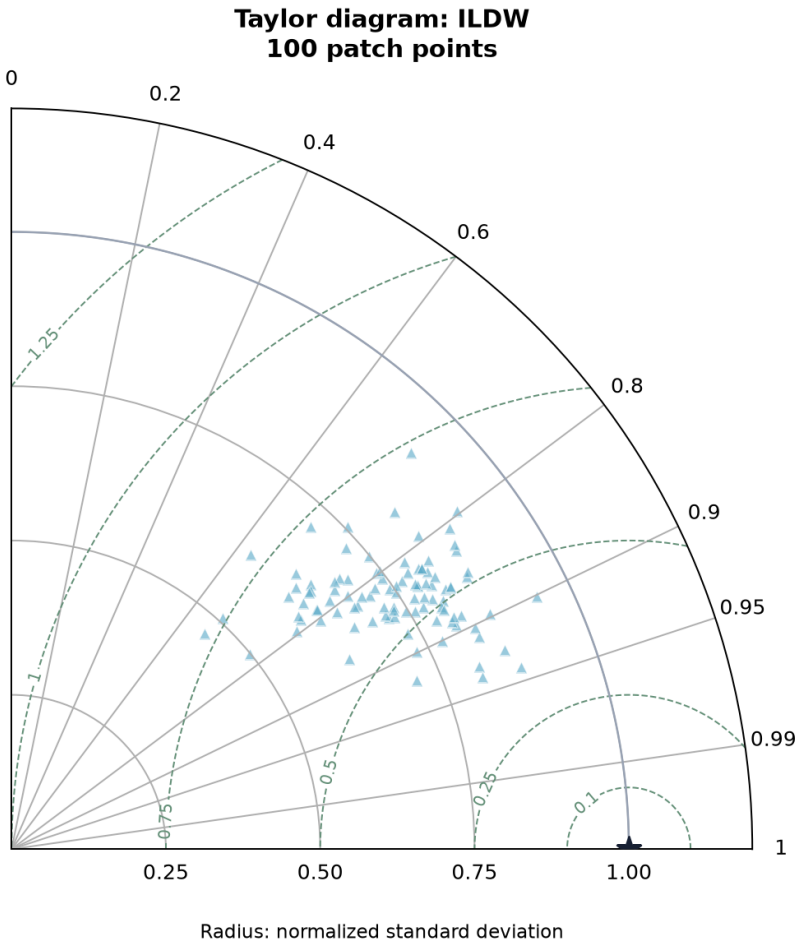
Homotopy Solver no shrinkage


Taylor diagram: IDW
100 patch points
0
0.2
0.4
0.6
0.8
0.9
0.95
0.99
1
1.25
1
0.75
0.5
0.25
0.1
0.25
0.50
0.75
1.00
Radius: normalized standard deviation

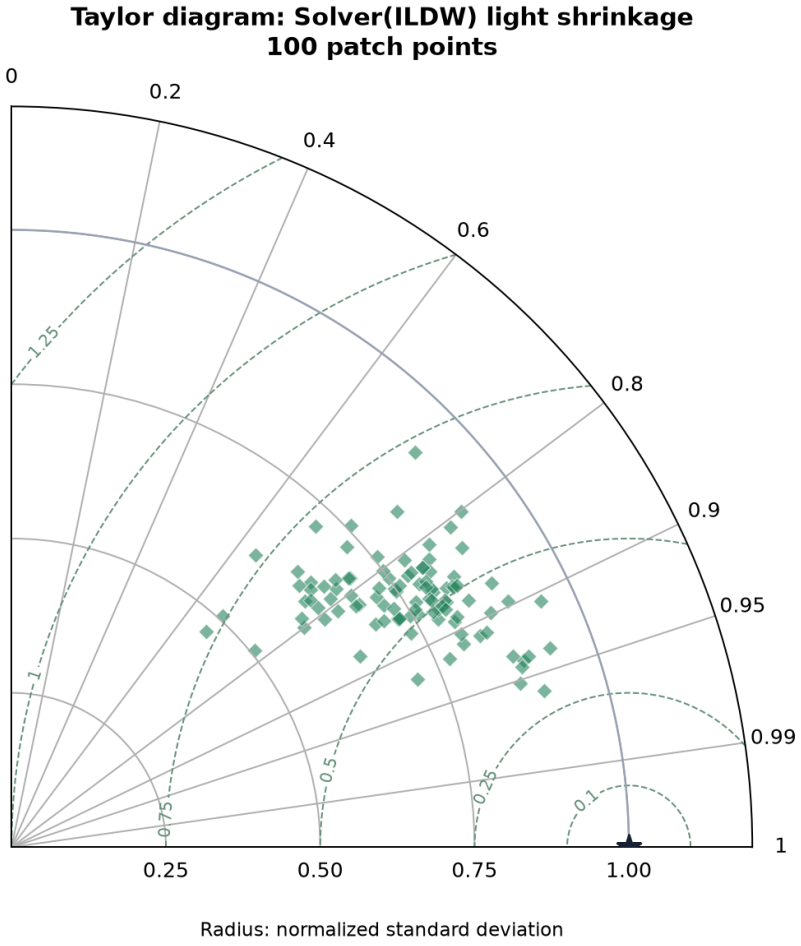
IDW



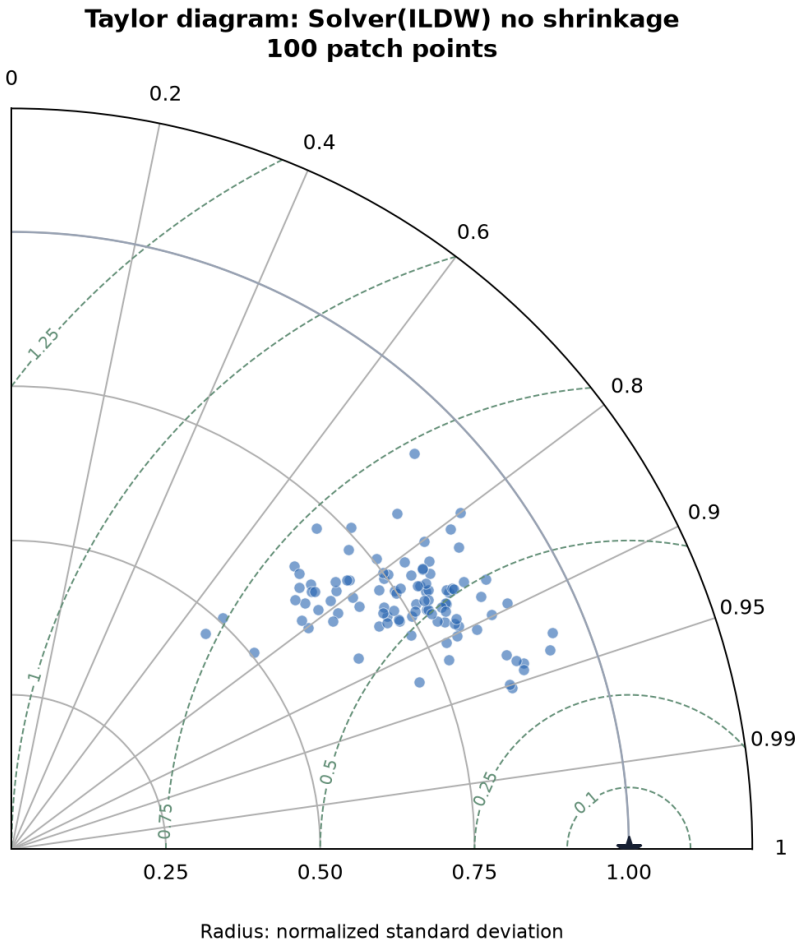
ILDW-init Roy DCT


Taylor diagram: ILDW
100 patch points
0
0.2
0.4
0.6
0.8
0.9
0.95
0.99
1
1.25
1
0.75
0.5
0.25
0.1
0.25
0.50
0.75
1.00
Radius: normalized standard deviation

ILDW



Solver ILDW light shrinkage



Solver ILDW no shrinkage

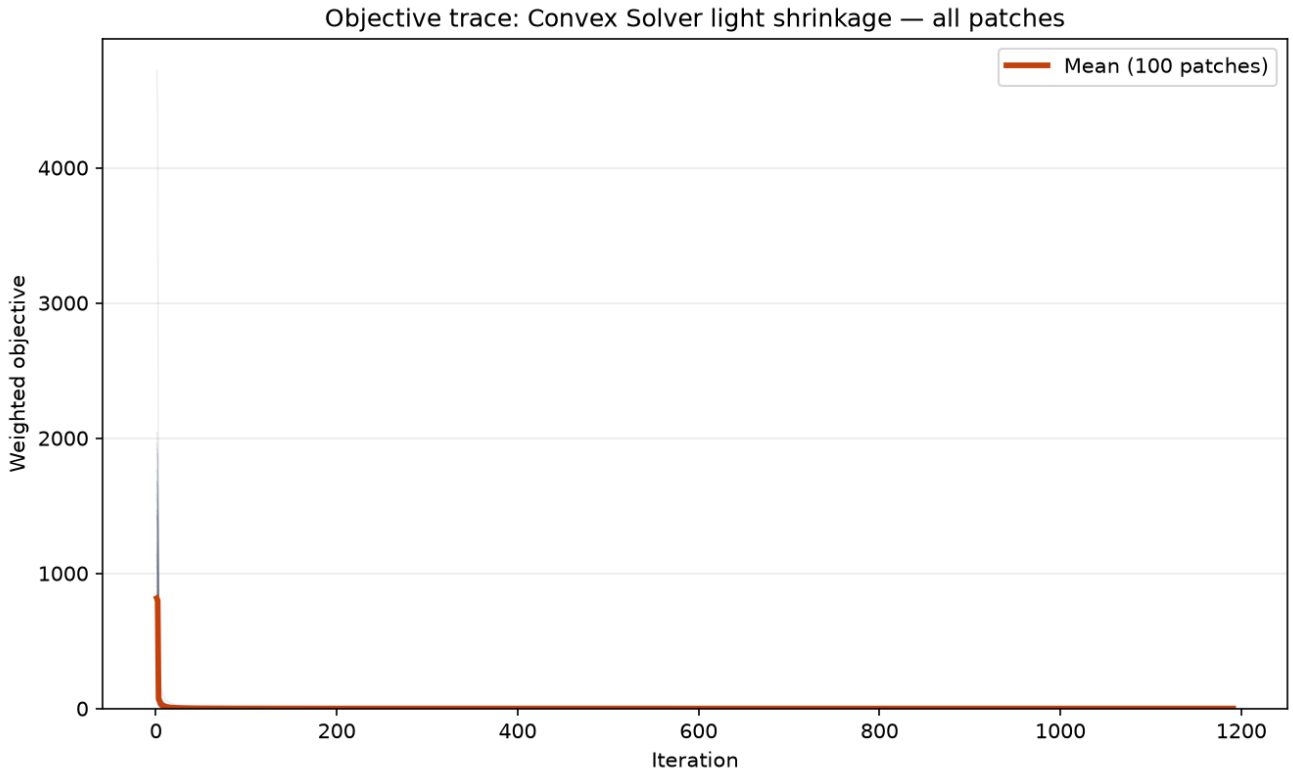
3. Wet/dry confusion summary

Method	Fraction of all pixels				Conditional rate			
	True wet	False wet	True dry	False dry	True wet	False wet	True dry	False dry
IDW	0.704	0.123	0.168	0.005	0.992	0.456	0.544	0.008
ILDW	0.703	0.114	0.177	0.006	0.991	0.423	0.577	0.009
Solver(ILDW) light shrinkage	0.703	0.111	0.180	0.006	0.991	0.405	0.595	0.009
Solver(ILDW) no shrinkage	0.703	0.111	0.180	0.006	0.991	0.405	0.595	0.009
Convex Solver light shrinkage	0.702	0.071	0.220	0.007	0.989	0.268	0.732	0.011
Convex Solver no shrinkage	0.702	0.071	0.220	0.007	0.989	0.268	0.732	0.011
Homotopy Solver light shrinkage	0.702	0.071	0.220	0.007	0.989	0.268	0.732	0.011
Homotopy Solver no shrinkage	0.702	0.071	0.220	0.007	0.989	0.268	0.732	0.011
ILDW-init Roy DCT	0.703	0.115	0.176	0.006	0.990	0.424	0.576	0.010

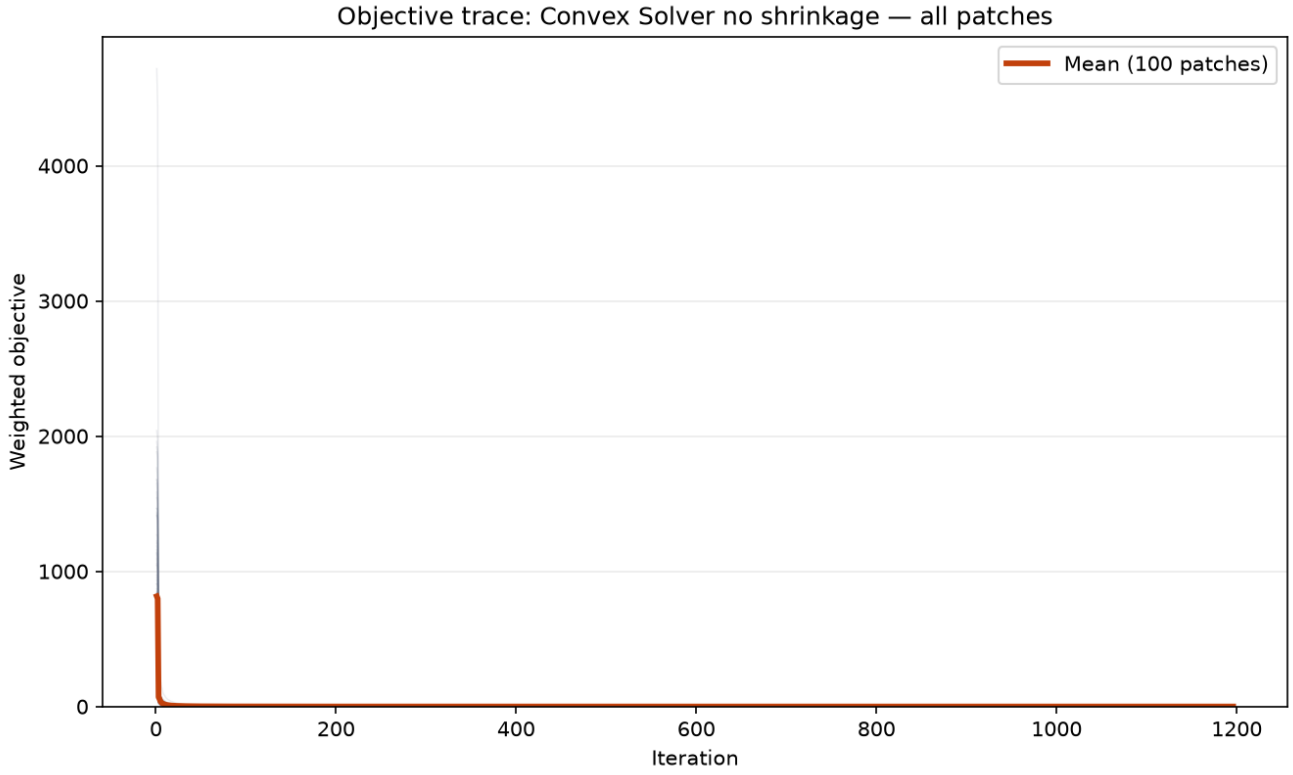
Patch-averaged wet/dry confusion summaries at a threshold of 0.6 mm/h (wet means rainfall greater than or equal to the threshold). In the fraction-of-all-pixels columns, each class is divided by the total number of pixels in each patch before averaging across 100 patches. In the conditional-rate columns, true wet and false dry are normalized by wet ground-truth pixels, while true dry and false wet are normalized by dry ground-truth pixels. Rates are calculated per patch and then averaged; patches with a zero denominator are excluded from the corresponding conditional rate. Values are rounded to three decimal places.

4. All-patch optimization behavior

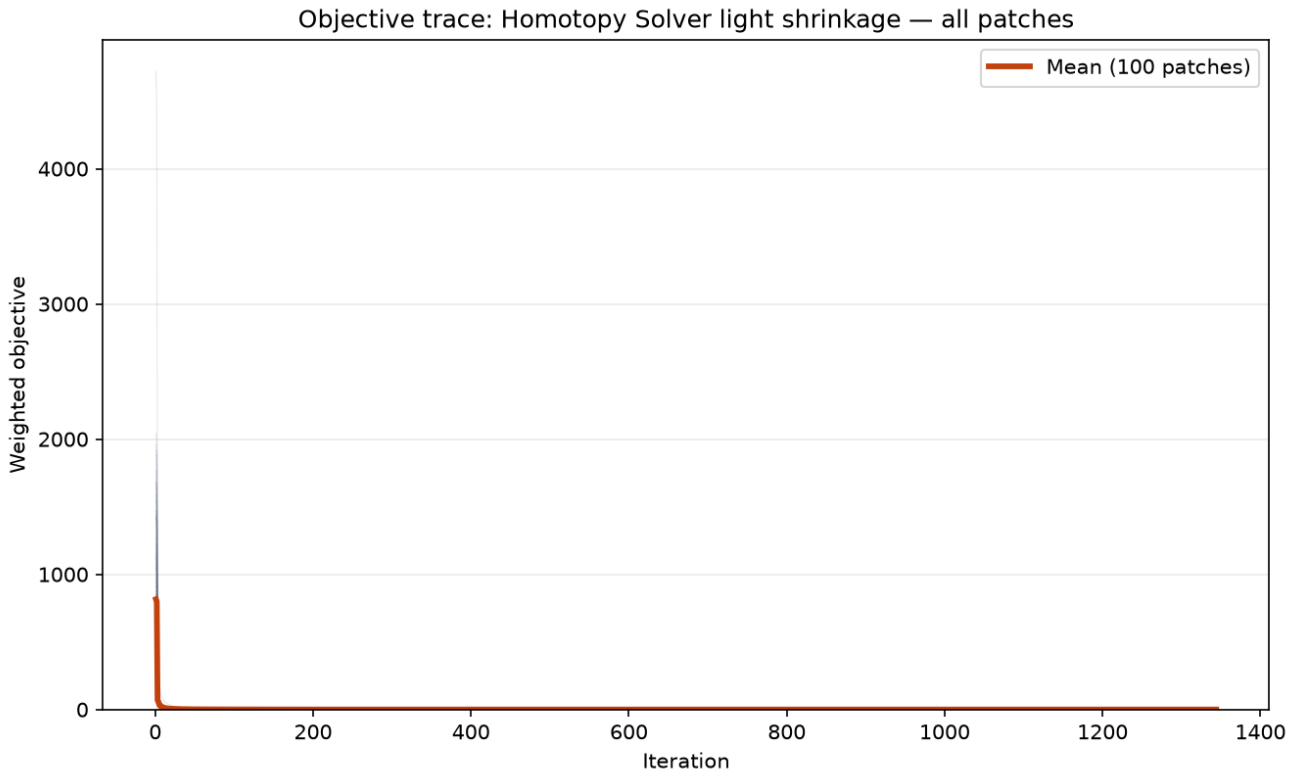
Faint lines are individual patches; the thick orange line is the arithmetic mean per iteration.



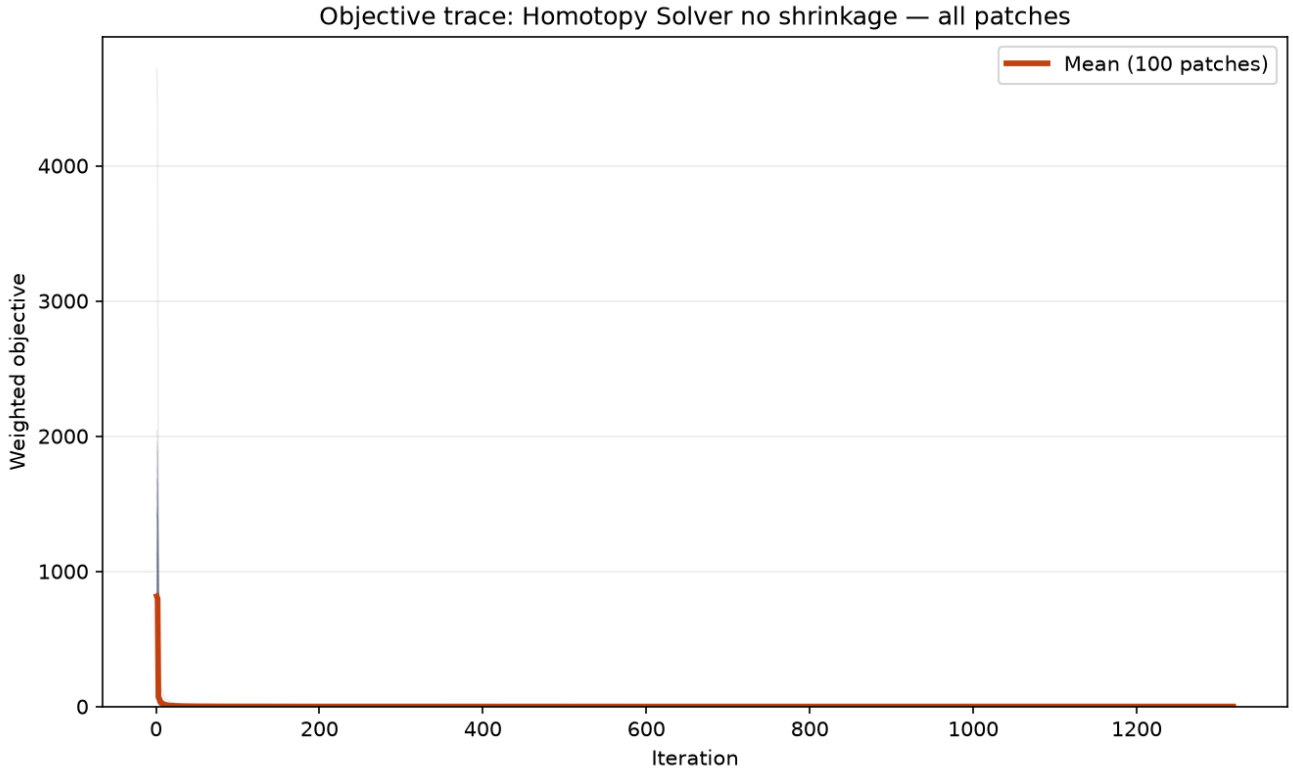
Convex Solver light shrinkage



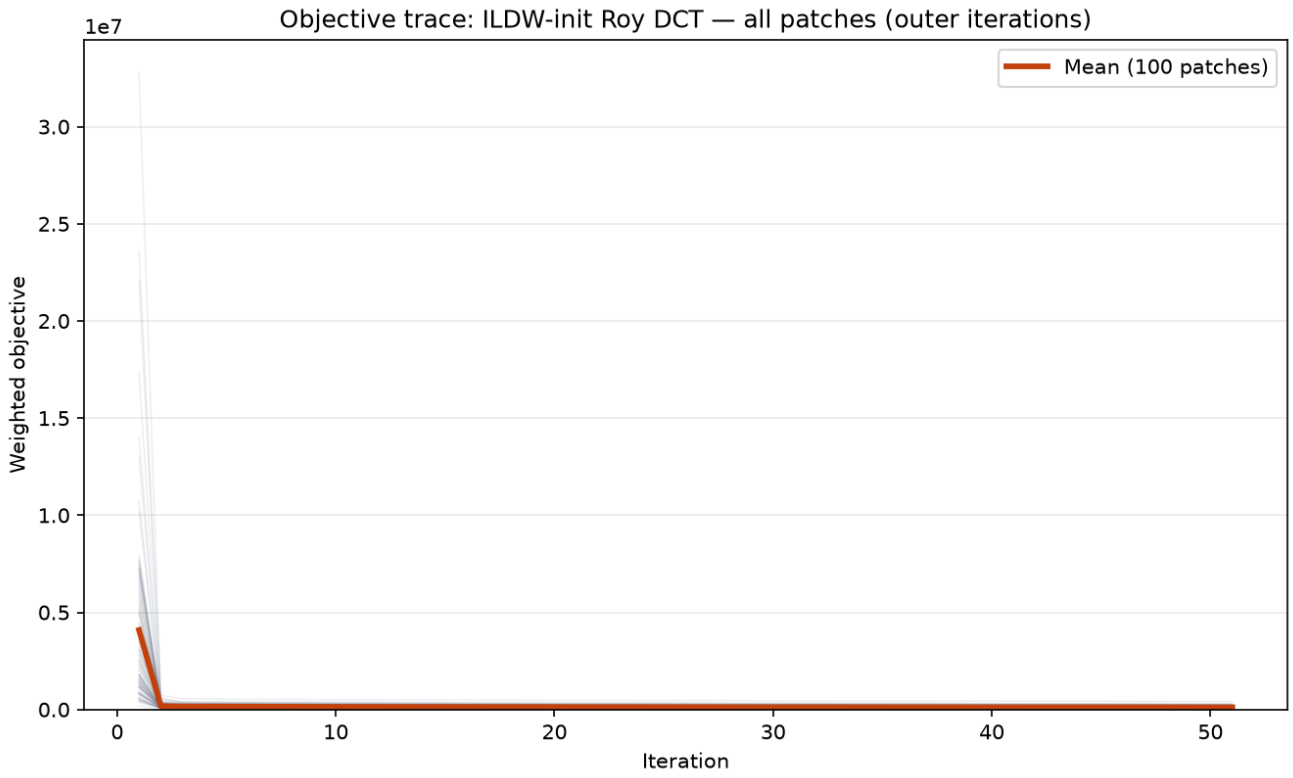
Convex Solver no shrinkage



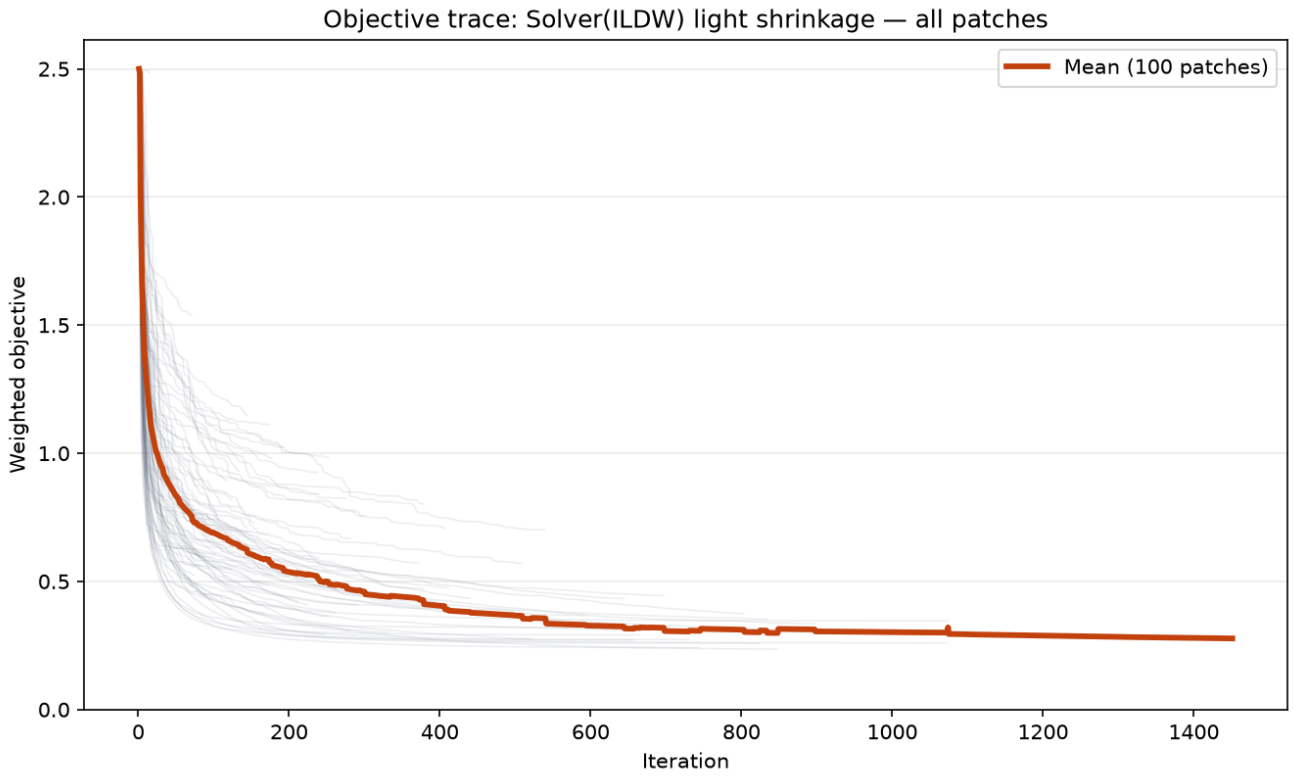
Homotopy Solver light shrinkage



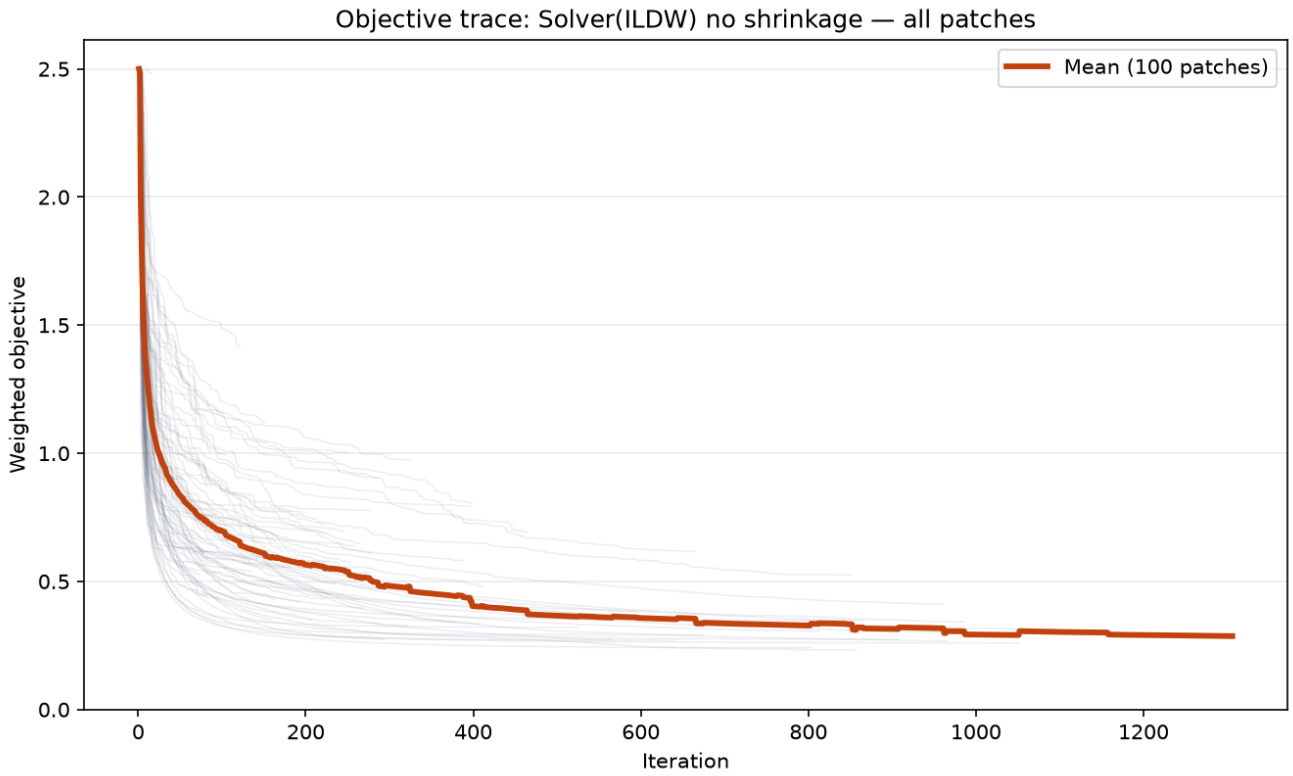
Homotopy Solver no shrinkage



ILDW-init Roy DCT



Solver ILDW light shrinkage



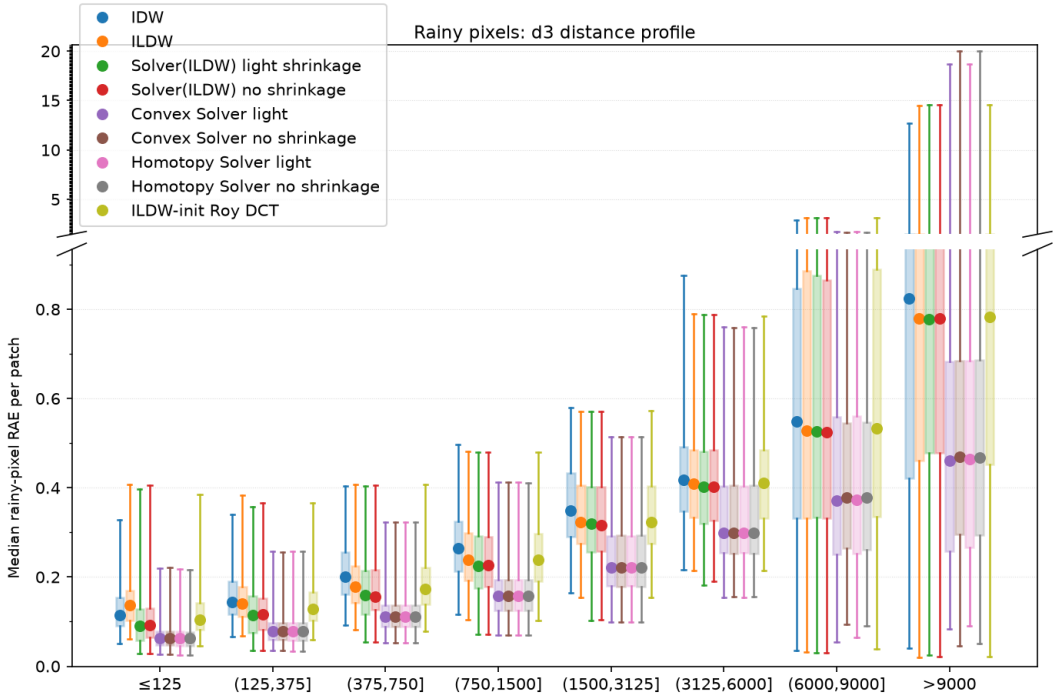
Solver ILDW no shrinkage

5. D3 distance summaries

Distance Profiles Box Whisker - d3 (third-closest link)



distance iqr medians nonrainy multi k3 box whisker



distance iqr medians rainy multi k3 box whisker

6. Solver runtimes

Solver directory	N	Mean s	Median s	Total h
sol_dir_convex_light_shrinkage	100	83.55	79.4	2.321
sol_dir_convex_no_shrinkage	100	79.96	73.71	2.221
ILDW-init Roy DCT	100	537.9	373.5	14.94
Solver(ILDW) no shrinkage	100	120.5	74.68	3.346
Solver(ILDW) light shrinkage	100	45.34	27.81	1.259